Simon Bunel

Contact Information

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Citizenship: French

Research Interests

Productivity, Innovation and Technological Change, Economics of AI, Regulation.

Employment

Postdoctoral Researcher, INSEAD, Stone Centre for the Study of Wealth Inequality (2026-).

Economist, Banque de France (2020-2025).

Economist, Insee, French Statistical Office (2017-2020).

Teaching

Adjunct Professor, École Normale Supérieure, Ulm (2022-).
Introduction to economic growth theories
Introduction to contemporary economic issues

Lecturer, ENSAE (2021-2022).
Introduction to contemporary economic issues

Education

Paris School of Economics, Paris.
Ph.D. in Economics, 2024. Advisor: Philippe Aghion.

Paris School of Economics, Paris.
Master in Analysis and Policy in Economics, 2017.

ENSAE, Paris.
Master, *Administrateur de l'Insee* training, 2016.

École polytechnique, Palaiseau.
Bachelor of Science (X2012), 2015.

Other Affiliations

Collège de France, Farhi Innovation Lab
Associate Researcher (2016-).

Presses Universitaires de France
Director of the "Economic Challenges" series, *Défis Économie* (2023-).

Job Market Paper

Lean Years for Productivity but Seeds of Innovation: Effects of Pesticide Regulation in French Agriculture
[Latest version]

Economists have long debated the impact of regulation on productivity. Identifying this impact is challenging because many confounding factors typically affect a firm's production function. This paper examines it in a setting with a simple production function: French crop farming. Specifically, I study how increasingly restrictive pesticide bans have affected farm productivity and agrochemical innovation. To this end, I combine French farm-level balance sheets and crop protection records with the registry of pesticide authorizations. I find that bans on inputs without substitutes are the most costly for farms, but also induce the most innovation, including entirely new active molecules. Regulation therefore involves a trade-off between short-term productivity losses and medium-term technological progress.

Publications

Modern Manufacturing Capital, Labor Demand, and Product Market Dynamics: Evidence from France, with Philippe Aghion, Céline Antonin, and Xavier Jaravel.
Conditionally accepted, *American Economic Journal: Macroeconomics* (2026).

How Different Uses of AI Shape Labor Demand: Evidence from France, with Philippe Aghion, Xavier Jaravel, Thomas Mikaelsen, Alexandra Roulet, and Jakob Søgaard.
AEA Papers and Proceedings (2025).

The Local Labor Market Effects of Modern Manufacturing Capital: Evidence from France, with Philippe Aghion, Céline Antonin, and Xavier Jaravel.
AEA Papers and Proceedings (2023).

Firm Dynamics and Growth Measurement in France, with Philippe Aghion, Antonin Bergeaud, and Timo Boppart.
Journal of the European Economic Association (2018).

Work in Progress

Social Spillovers and AI Adoption: Evidence from Denmark, with Philippe Aghion, Xavier Jaravel, Thomas Mikaelsen, Alexandra Roulet, and Jakob Søgaard.

The Returns to Competitive Industrial Policy: Evidence from France's Mission-Oriented Programs, with Philippe Aghion, Florian El Gourdou, Xavier Jaravel, and Alexandra Roulet.

Economic Growth and Ageing: Evidence from France, with Kai Arvai and Sebastian Stumpner.

Books and Chapters

The Power of Creative Destruction, with Philippe Aghion and Céline Antonin.
Harvard University Press (2021).

The Direct and Indirect Effects of Automation on Employment: A Survey of the Recent Literature, with Philippe Aghion, Céline Antonin, and Xavier Jaravel.
In *Robots and AI: A New Economic Era* (2022), Lili Yan Ing and Gene M. Grossman (eds.), Oxon and New York: Routledge.

Policy Reports and Other Publications

Labour Market Policies in the Age of AI, with Philippe Aghion, Antonin Bergeaud, and Alexandra Roulet.
CEPR Europe 2050 Paper (2026).

AI and Growth: Where Do We Stand? with Philippe Aghion.
Policy Note (2024).

AI: Our Ambition for France, with the French Artificial Intelligence Commission.
Policy report for the French President (2024).

Digitalisation and Productivity, with the European Central Bank expert group on productivity.
ECB Occasional Paper, No. 2024/339 (2024).

An Evaluation of the French Innovation Tax Credit, with Benjamin Hadjibeyli.
Economics and Statistics (2021).

Artificial Intelligence, Growth and Employment: The Role of Policy, with Philippe Aghion and Céline Antonin.
Economics and Statistics (2019).

Publications in French

IA générative : un gel des embauches de jeunes dans les métiers exposés, with Marco Guido Palladino, Antonin Bergeaud, Antoine Bertheau, and Oscar Degos.
LIFT Policy Paper and *LIFT Working Paper* (2026).

Pourquoi l'IA ne rend pas (encore) la France plus productive, with Antonin Bergeaud.
LIFT Policy Paper (2026).

Comment lutter contre le décrochage de la productivité, with Antonin Bergeaud, Xavier Jaravel, and Alexandra Roulet.
Note du Conseil d'analyse économique (2026).

Des PIA à France 2030 : évaluation des aides compétitives en faveur de la croissance des entreprises, with Philippe Aghion, Florian El Gourdou, Xavier Jaravel, and Alexandra Roulet.
Focus du CAE (2026).

Diffusion de l'intelligence artificielle et dépendance numérique : fondements de l'intervention publique et enjeux de mesure, with Alice Lapeyre and Antoine Lopes.
Focus du CAE (2026).

Les incitations fiscales à la R&D et l'innovation : état des lieux, effets et alternatives, with Michaël Sicsic.
In *Les politiques publiques par la défiscalisation* (2025), C. Carbonnier, N. Morel, B. Palier, and M. Zemmour (eds.), Presses de Sciences Po. Earlier version: *LIEPP Working Paper* (2024).

Réexamen de l'écart de performance de l'Europe vis-à-vis des États-Unis, with Alexander Clymo, Olivier Garnier, and Riccardo Zago.
Bloc-notes Éco, Banque de France (2025).

IA : Notre ambition pour la France, with the Commission de l'Intelligence artificielle.
Rapport pour le Président de la République (2024).

Aides à la R&D : pratiques internationales et revue de la littérature sur leurs effets, with Michaël Sicsic.
Revue de l'OFCE (2021).

Évaluation du dispositif Jeune entreprise innovante. Un exemple d'application du modèle d'analyse de sensibilité de Rosenbaum, with Simon Quantin and Clémence Lenoir.
Document de travail Insee (2021). Short version: *Insee Analyses*, no. 68.

Évaluation du crédit d'impôt innovation : dynamique des bénéficiaires depuis son introduction, with Benjamin Hadjibeyli.
Insee Références - Les entreprises en France, pp. 73-84 (2019).

Seminars and Conferences

2026 INRAE SMART (Rennes); INSEAD - Collège de France Brown Bag Seminar (Paris); ECO Brown Bag Seminar (Paris); Paris-Lyon Public Economics Workshop (Lyon); BSE Summer Forum *Economics of Science and Innovation* (Barcelona); CEPR RPN Growth, Innovation and Social Model in Europe, Inaugural Workshop (Collège de France, Paris); AI and Society Conference (Bocconi, Milan); Agrochemicals: health, globalization, and biodiversity PSE Workshop (Paris); Zurich Workshop in AI & Applied Economics (Zurich); CREST (Palaiseau); INSEAD (forthcoming); CEPR Paris Symposium (forthcoming).

2025 ASSA (San Francisco); Tech for Finance: AI & Blockchain (Dauphine, Paris); Banque de France / Fondation France Japon Lab Spring Workshop (Paris); Workshop *The Economics of Artificial Intelligence* (Collège de France, Paris); LAGV (Marseille); EEA (Bordeaux); AI and Economics PSE Workshop (Paris); Le Mans University; Banque de France.

2023 CEPR ESSIM (Tarragona); PILLARS Workshop: Digitization, AI and Labor Outcomes (Paris).

2022 University of Bern; BSE Summer Forum *Firms in the Global Economy* (Barcelona); BSE Summer Forum *Economics of Science and Innovation* (Barcelona); EEA (Milan); ESCB Cluster 2 (Athens).

2021 Banque de France (virtual); PSE Macro PhD Workshop (virtual); AFSE (virtual); SSES (virtual); LAGV (Marseille); IIPF (virtual); EEA (virtual); EARIE (virtual); EALE (virtual); NBER Doctoral Student Workshop on Economics of AI (virtual); CompNet Annual Conference (Paris); TEPP (Évry); OECD (virtual); Conference on Firm Financing, Organization and Dynamics (virtual).

2020 GREThA Seminar (Bordeaux); NBER Summer Institute (virtual, as panelist); EEA (virtual); Workshop Collège de France - Banque de France (virtual); AFSE-Trésor (virtual).

2019 AFSE (Orléans); IAAE (Nicosia); EARIE (Barcelona); Workshop Collège de France - CREST (Paris); AFSE-Trésor (Paris).

2018 PEJ Annual Meeting (Lisbon).

2017 AFSE (Nice).

Discussions

1st Dual-Venue Transatlantic Workshop in Energy and Nature Finance, May 2026.
"The Disclosure Channel of Knowledge Spillovers: Evidence from Environmental Innovation" by Leo Liu, Elvira Sojli, Wing Wah Tham, and Cara Vansteenkiste.

AI, Digitization and Trade Patterns PSE Workshop, December 2025.
"Automation without Borders: The Impacts of Generative AI on International Service Outsourcing" by Neha Betai and Maggie Chen.

PSE-CEPR Policy Forum, June 2025.
"Forecasting Generative AI Work Capabilities with a LLM Dynamic Benchmark" by R. Maria del Rio-Chanona and Johanna Einsiedler.

CEPR ESSIM, May 2024.
"Artificial Intelligence and Jobs: Evidence from US Commuting Zones" by Alessandra Bonfiglioli, Rosario Crinò, Gino Gancia, and Ioannis Papadakis.

HEC Paris Entrepreneurship Workshop, December 2023.
"The Impact of Cloud Computing and AI on Industry Dynamics and Competition" by Yao Lu, Gordon M. Phillips, and Jia Yang.

CFM-Warwick-Vienna Macro Conference, March 2023.
"How Costly Are Cartels?" by Flavien Moreau and Ludovic Panon.

CompNet Annual Conference, September 2022.
"Benefits and Costs of the ETS in the EU, a Lesson Learned for the CBAM Design" by Justus Böning, Virginia Di Nino, and Till Folger.

CompNet ProdTalk, May 2022.
"Robots, Firms, and Workforce Composition" by Liuchun Deng, Steffen Müller, Verena Plümpe, and Jens Stegmaier.

Refereeing

Quarterly Journal of Economics, *American Economic Journal: Applied Economics*, *American Economic Journal: Economic Policy*, *American Economic Review: Insights*, *Annals of Economics and Statistics*, *Applied Economics Letters*, *Applied Financial Economics*, *Economic Journal*, *Economica*, *Economics and Statistics*, *Economics of Transition*, *European Economic Review*, *German Economic Review*, *Industry and Innovation*, *IZA World of Labor*, *Journal of Human Resources*, *Journal of Labor Economics*, *Journal of the European Economic Association*, *Labour Economics*, *PLOS One*, *Regional Science and Urban Economics*, *Research Policy*, *Review of Economics and Statistics*, *World Development*.

Languages

French	Native.
English	Fluent.
Spanish	Basic knowledge.

Programming Skills

Software	R, Stata, SAS.

References

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Alexandra Roulet
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